

TIMOTIUS WILLIAM WUISANG

Engineering Physics Fresh Graduate | R&D & Experimental Research | Data Analytics | IoT & Instrumentation | Building & Energy Systems
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PROFESSIONAL SUMMARY

Engineering Physics fresh graduate from Universitas Gadjah Mada (GPA 3.57/4.00) with hands-on experience in materials R&D, IoT-based building systems, Python data analysis, instrumentation, and technical education. Evaluated thermochromic transparent wood across 3 coating concentrations and 4 key performance metrics, developed an automated ventilation concept in an industrial R&D environment, and built a machine-learning workflow achieving 69% classification accuracy. Experienced in translating experiments and quantitative data into structured technical documentation, recommendations, and presentations.

EDUCATION

Bachelor of Engineering in Engineering Physics | Universitas Gadjah Mada | Yogyakarta 2022 - 2026
• GPA: 3.57/4.00 | Focus areas: building physics & energy efficiency, instrumentation & control, data analytics, IoT/embedded systems, and material characterization.

RELEVANT EXPERIENCE

Research Assistant - Thermochromic Transparent Wood Project | Universitas Gadjah Mada | Yogyakarta Feb - Jul 2026
• Executed multi-stage materials research from delignification and polymer impregnation through polyurethane/thermochromic preparation and spray coating across 3 pigment concentrations (5%, 10%, and 15%).
• Characterized 4 key performance areas - CIELAB color response, luminous transmittance, response time, and thermal conductivity - and analyzed repeated measurements to compare formulation trade-offs.
• Quantified response-time trends of 286.81 +/- 18.28 s (5%), 332.40 +/- 4.95 s (10%), and 343.00 +/- 2.83 s (15%); analysis supported 10% as the most balanced formulation for visible color response and optical/thermal performance.
• Prepared literature reviews, technical documentation, data visualizations, and research summaries to support evaluation and reporting.

Research Engineer Intern - Research Division | YKK AP R&D Center Indonesia | Tangerang Dec 2024 - Jan 2025
• Developed and tested a preliminary IoT-based automated window-ventilation concept integrating environmental sensors, microcontroller/RTC control, and actuator to support passive-building strategies.
• Implemented time- and temperature-based control logic, including a night-ventilation operating window of 18:00-06:00, and analyzed indoor/outdoor environmental measurements to evaluate thermal-condition changes.
• Collaborated in industrial R&D discussions and co-authored a research proposal, translating experimental findings into practical engineering recommendations for building energy efficiency.

Online Mathematics Tutor | Independent / Self-Employed | Remote Sep 2025 - Present
• Provide recurring one-on-one mathematics tutoring totaling approximately 4 hours/week, adapting explanations and practice to student learning gaps and pace.
• Teach arithmetic, ratios, fractions, algebra, geometry, exponents/radicals, and competition-style problems using step-by-step reasoning, strengthening technical communication and mentoring capability.

Basic Informatics Laboratory Assistant | Universitas Gadjah Mada | Yogyakarta Aug - Dec 2024
• Supported Python practicum sessions across one academic semester, explaining programming fundamentals, troubleshooting coding issues, and adapting guidance to different student skill levels.
• Evaluated assignments and practicum reports and delivered structured feedback on programming logic, code correctness, and report clarity while coordinating with the teaching team.

AI Project Participant | Intel Prakarsa Muda 2021 - Sagasitas Indonesia | Yogyakarta Jun - Nov 2021
• Completed a 6-month team-based AI technology project program, contributing to concept development and communicating the project approach to mentors.

SELECTED PROJECTS

Conveyor Monitoring & Safety Automation System | Raspberry Pi, Sensors, Web Monitoring - Developed real-time monitoring for 6 operating parameters (vibration, voltage, current, power, load/mass, and brightness) with remote ON/OFF control, manual override, and automatic shutdown logic for abnormal conditions including current >3 A or voltage >13 V.

E-commerce Shipping Analytics | Python, Pandas, Matplotlib, Scikit-learn - Performed exploratory analysis and built a classification workflow for shipment prediction, achieving 69% accuracy. [Kaggle](#)

Process Engineering Capstone - Distillation & Instrumentation | DWSIM - Supported distillation-section design, condenser/reboiler review, instrument specification, process-control requirements, and basic control-valve sizing.

TRAINING

Data Science Training | KMTETI FT UGM Workshop Division | Yogyakarta Sep - Nov 2023
• Completed a 3-month introductory program covering data processing, visualization, and machine-learning workflow fundamentals.

SKILLS

Engineering & Research: Experimental testing, material characterization, heat transfer, building physics, instrumentation, technical documentation, laboratory troubleshooting

Embedded & Automation: Arduino, ESP32, STM32, Raspberry Pi, sensor integration, actuator control, circuit/system troubleshooting

Data & Digital: Python, Pandas, Matplotlib, Scikit-learn, Microsoft Excel, data cleaning, exploratory analysis, data visualization, DWSIM

Professional: Analytical problem solving, teamwork, technical communication, mentoring, presentation, adaptability, willingness to learn

Languages: Indonesian (Native), English (Intermediate)